

3.6 Carboxylic Acids & Derivatives

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.6 Carboxylic Acids & Derivatives
Difficulty	Medium

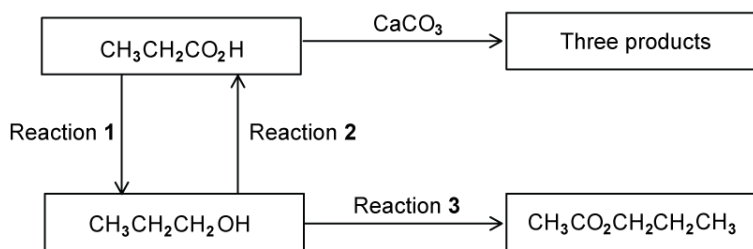
Time allowed: 60

Score: /46

Percentage: /100

Question 1a

A series of reactions based on propanoic acid is shown.



Write an equation for reaction 1, using [H] to represent the reducing agent.

[2 marks]

Question 1b

i)

What type of reaction is reaction 2?

[1]

ii)

Suggest a suitable reagent and conditions for reaction 2.

[2]

[3 marks]

Question 1c

Write an equation for the reaction of propanoic acid with calcium carbonate, CaCO_3 .

[2 marks]

Question 1d

i)

Suggest a suitable reagent and conditions for reaction 3.

[2]

ii)

Identify the other product of reaction 3.

[1]

[3 marks]**Question 2a**An organic ester, **B**, has the empirical formula C_2H_4O . An experiment by a student in a college gave a value of 87.5 for M_r of **B**.What is the molecular formula of **B**?**[1 mark]****Question 2b**In the boxes below, draw the structural formulae of four isomers of **B** that are esters.

W	X
Y	Z

[4 marks]

Question 2c

The student hydrolysed his sample of **B** by heating with aqueous mineral acid and then separating the alcohol, **C**, that was formed. He heated the alcohol **C** under reflux with acidified dichromate(VI) ions and collected the product **D**.

A sample of **D** gave an orange precipitate with 2,4-dinitrophenylhydrazine reagent. A second sample of **D** gave no reaction with Tollens' reagent.

i)
What group does the reaction with 2,4-dinitrophenylhydrazine reagent show to be present in **D**?

[1]

ii)
What does the result of the test with Tollens' reagent show about **D**?

[1]

iii)
What is the structural formula of the alcohol **C**?

[1]

iv)
Which of your esters, **W**, **X**, **Y**, or **Z** has the same structure as that of the ester **B**?

[1]

[4 marks]

Question 2d

Which, if any of your esters, **W**, **X**, **Y**, or **Z** is chiral? Explain your answer.

[1 mark]

Question 3a

Esters are compounds which provide the flavour of many fruits and the perfumes of many flowers.

The ester $\text{CH}_3(\text{CH}_2)_2\text{CO}_2\text{CH}_3$ contributes to the aroma of apples.

i)

State the reagents and conditions needed for the hydrolysis of this ester.

[2]

ii)

Write the equation for the hydrolysis of this ester.

[1]

iii)

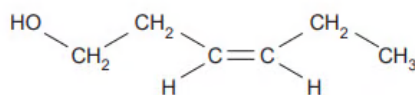
Apart from their use as perfumes and food flavourings, state one major commercial use of esters.

[1]

[4 marks]

Question 3b

Leaf alcohol is a stereoisomer that can form when insects such as caterpillars eat green leaves.



i)

Draw the other stereo-isomer of leaf alcohol.

[1]

ii)

Draw the structure for the ester formed when leaf alcohol reacts with ethanoic acid.

Show all the bonds in the ester group

[2]

[3 marks]

Question 3c

i)

Deduce the relative molecular mass, M_r , for leaf alcohol.

[1]

ii)

Leaf alcohol was reacted to form a product with an M_r value 18 units less.

Suggest a structure for this product and deduce the type of reaction that took place

structure of product

type of reaction

[2]

[3 marks]

Question 3d

Describe a simple chemical test to distinguish between leaf alcohol and your product in **part (c)(ii)**.

test

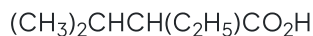
observation

[2 marks]

Question 4a

Fermentation of sugars by bacteria or moulds produces many different organic compounds.

One compound present in fermented molasses is 2-ethyl-3-methylbutanoic acid which gives a distinctive aroma to rum.



2-ethyl-3-methylbutanoic acid

i)

What is the molecular formula of 2-ethyl-3-methylbutanoic acid?

[1]

ii)

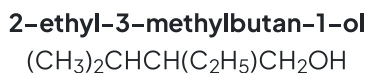
How many chiral carbon atoms are present in a molecule of 2-ethyl-3-methylbutanoic acid? If none write 'none'.

[1]

[2 marks]

Question 4b

A sample of 2-ethyl-3-methylbutanoic acid may be prepared in a school or college laboratory by the oxidation of:



i)

State the reagent(s) that would be used for this oxidation.

Describe what colour change would be seen.

reagent(s)

colour change from to

[3]

ii)

This reaction is carried out by heating the reacting chemicals together.

What could be the main organic impurity present in the sample of the acid?

Explain your answer.

[2]

iii)

State whether a distillation apparatus or a reflux apparatus should be used.

Explain your answer.

[1]

[6 marks]

Question 4c

A structural isomer of 2-ethyl-3-methylbutan-1-ol is 2-ethyl-3-methylbutan-2-ol, $(\text{CH}_3)_2\text{CHC}(\text{OH})(\text{C}_2\text{H}_5)\text{CH}_3$.

What colour change would be seen if this were heated with the reagents you have given in part **(b)(i)**?

Explain your answer as clearly as you can.

[3 marks]

Question 4d

An isomer of 2-ethyl-3-methylbutanoic acid which is an ethyl ester is a very strong smelling compound which is found in some wines. This ethyl ester contains a branched hydrocarbon chain and is chiral.

Draw the displayed formula of this ethyl ester.

Identify the chiral carbon atom with an asterisk (*).

[3 marks]

Model Answers: Medium

1a

a) An equation for reaction 1, using [H] to represent the reducing agent is:



- Correct reactants; [1 mark]
- Correct products; [1 mark]

[Total: 2 marks]

- Reaction 1 shows the reduction of propanoic acid
- The reduction of this alcohol will produce propanol and water

1b

b)

i) Reaction 2 is:

- Oxidation; [1 mark]

ii) A suitable reagent and conditions for reaction 2 is:

- Acidified potassium dichromate (VI) $\text{K}_2\text{Cr}_2\text{O}_7$ / acidified potassium manganate (VII) / KMnO_4 ; [1 mark]
- Reflux; [1 mark]

[Total: 3 marks]

- You can see that oxygen has been gained in the formula for the propanoic acid compared to propanol
- Acidified potassium dichromate (VI) and acidified potassium manganate (VII) are common oxidising agents you should be aware of
- They will be reduced as they oxidise the propanol

1c

c) An equation for the reaction of propanoic acid with calcium carbonate, CaCO_3 is:



- Correct equation; [1 mark]
- Correctly balanced; [1 mark]

[Total: 2 marks]

- Propanoic acid reacts like any other acid
 - Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide
 - The salt formed is calcium propanoate

1d

d)

i) A suitable reagent and conditions for reaction **3** is:

- $\text{CH}_3\text{CO}_2\text{H}$; [1 mark]
- Warm / hot / higher temperature / heat / reflux **AND** concentrated sulfuric acid; [1 mark]

ii) The other product of reaction **3** is:

- Water; [1 mark]

[Total: 3 marks]

- Reaction **3** shows propanol forming propyl ethanoate, an ester
- An ester is formed when an alcohol reacts with a carboxylic acid
- The first part of the name comes from the alcohol, and the second part from the carboxylic acid
- To form propyl ethanoate, ethanoic acid is therefore the reagent needed
- Esterification reactions are known as condensation reactions as they lose a molecule of water when the ester linkage forms

2a

a) The molecular formula of **B** is:

- $C_4H_8O_2$; [1 mark]

[Total: 1 mark]

- The M_r of the empirical formula is $(2 \times 12) + 4 + 16 = 44$
- $87.5 / 44 = 1.99$
- You therefore multiply the empirical formula by 2

2b

b) The structural formulae for the isomers are:

- **W** = $HCO_2CH(CH_3)_2$; [1 mark]
- **X** = $HCO_2CH_2CH_2CH_3$; [1 mark]
- **Y** = $CH_3CO_2CH_2CH_3$
OR
 $CH_3CO_2C_2H_5$; [1 mark]
- **Z** = $CH_3CH_2CO_2CH_3$
OR
 $C_2H_5CO_2CH_3$; [1 mark]

[Total: 4 marks]

- It doesn't matter which structural formulae you have given for each letter, each correct structure is worth one mark
- Start by drawing the ester functional group, $-COO-$ and then trying different lengths of carbon chain on either side until you have four combinations

2c

c)

i) The reaction with 2,4-dinitrophenylhydrazine reagent shows the presence of:

- $C=O$ group / carbonyl group; [1 mark]

ii) The result of the test with Tollens' reagent shows:

- There is no $-CHO$ / aldehyde group

OR

There is a ketone present; [1 mark]

iii) The structural formula of the alcohol **C** is:

- $(\text{CH}_3)_2\text{CHOH}$; [1 mark]

iv) The esters, **W**, **X**, **Y**, or **Z** has the same structure as that of the ester **B**

- Ester **W** / $\text{HCO}_2\text{CH}(\text{CH}_3)_2$; [1 mark]

[Total: 4 marks]

- Ester **B** has **four** carbon atoms in total from a combination of the carboxylic acid and alcohol
- The alcohol used to form Ester **B** has been oxidised to a ketone (shown by the lack of a positive result with Tollen's reagent)
- Ketones have an alkyl group bonded to either side of the $\text{C}=\text{O}$ bond
- The ketone formed must be propanone as one less carbon atom and it would be an aldehyde, and one more carbon atom and there would not be enough once it has formed an alcohol to form Ester **B**
- A secondary alcohol forms a ketone when it is oxidised so the alcohol must be propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$
- When looking at the structure of an ester
 - The chain attached to the oxygen atom (on the right hand side) in the $-\text{COO}-$ is from the alcohol
 - The chain attached to the carbon atom (on the left hand side) in the $-\text{COO}-$ is from the carboxylic acid
- Look at the structures drawn in part b)
 - You can narrow Ester **B** down to **W** and **X** as they both have three carbon atoms on the right hand side of the $-\text{COO}-$ group
 - Propan-1-ol would form Ester **X**
 - Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$ would form Ester **W**
- Your answer to part (iv) must match your drawing as you may not have drawn $\text{HCO}_2\text{CH}(\text{CH}_3)_2$ as Ester **W**

2d

d) Which, if any of your esters, **W**, **X**, **Y**, or **Z** is chiral:

- None

AND

Because there are no chiral centres present in any of the four esters; [1 mark]

- Propan-1-ol would form Ester **X**
- Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$ would form Ester **W**
- Your answer to part (iv) must match your drawing as you may not have drawn $\text{HCO}_2\text{CH}(\text{CH}_3)_2$ as Ester **W**

2d

d) Which, if any of your esters, **W**, **X**, **Y**, or **Z** is chiral:

- None

AND

Because there are no chiral centres present in any of the four esters; [1 mark]

[Total: 1 mark]

- A chiral centre is one in which a carbon atom is bonded to four different atoms / groups of atoms

3a

a)

i) The reagents and conditions needed for the hydrolysis of this ester are:

- Dilute Acid / sulfuric acid

OR

Dilute alkali / sodium hydroxide; [1 mark]

- Heat / reflux / warm; [1 mark]

ii) The equation for the hydrolysis of this ester

- $\text{CH}_3(\text{CH}_2)_2\text{CO}_2\text{CH}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_3(\text{CH}_2)_2\text{CO}_2\text{H} + \text{CH}_3\text{OH}$; [1 mark]

iii) One major commercial use of esters is:

Any **one** from:

- Solvents; [1 mark]
- Plastics; [1 mark]
- Textiles; [1 mark]

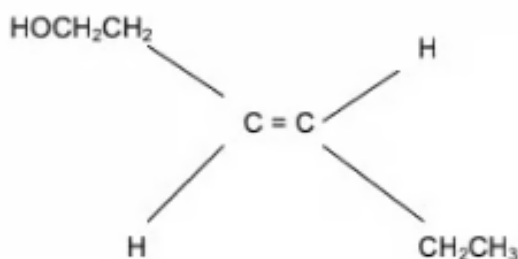
[Total: 4 marks]

- **Remember:** When an ester is hydrolysed, it reforms the carboxylic acid and alcohol it was made from
 - The left hand side of the structural formula is formed from the carboxylic acid:
 $\text{CH}_3(\text{CH}_2)_2\text{CO}_2\text{CH}_3$
 - There are four carbon atoms so butanoic acid was used
 - The right hand side of the structural formula is from the alcohol:
 $\text{CH}_3(\text{CH}_2)_2\text{CO}_2\text{CH}_3$
 - There is one carbon atom so methanol was used

3b

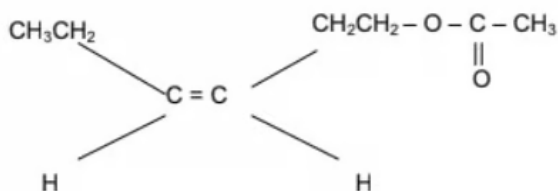
b)

i) The other stereoisomer of leaf alcohol is:



- Correct structure; [1 mark]

ii) The structure for the ester formed when leaf alcohol reacts with ethanoic acid is:



- Ester link; [1 mark]
- Rest of molecule; [1 mark]

[Total: 3 marks]

- To draw the other stereoisomer, just swap the position of the CH_2CH_3 group so that it is on the opposite side of the carbon carbon double bond
 - Cis isomers have their functional groups on the same side of the double bond
 - Trans isomers have their functional groups on opposite sides of the double bond
- Drawing the ester formed might look a bit tricky initially- but all you need to do is replace the $-\text{OH}$ group on leaf ethanol with an ester linkage and add CH_3 onto the other side

3c

c)

i) The relative molecular mass, M_r , for leaf alcohol is:

- 100; [1 mark]

ii) The structure of the product and type of reaction is:

- $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH} - \text{CH}_2\text{CH}_3$; [1 mark]
- Dehydration / elimination; [1 mark]

[Total: 3 marks]

- It would be easy to miscalculate the M_r for alcohol so count the number of atoms of each element first:
 - $(12 \times 6) + 16 + (1 \times 12) = 100$
- The command word suggest means you need to apply your understanding to a new context
 - You should recognise that water has an M_r of 18 and deduce that this has been lost from the alcohol
 - Water is lost from an alcohol to form an alkene
 - The double bond will form between the carbon atoms to which the OH group was attached

3d

d) A simple chemical test to distinguish between leaf alcohol and your product in part

(c)(ii) is:

Any pair from:

Alternative 1:

- Add PCl_5 ; [1 mark]
- Misty fumes; [1 mark]

Alternative 2:

- Add $\text{H}^+ / \text{Cr}_2\text{O}_7^{2-}$; [1 mark]
- Turns from orange to green; [1 mark]

Alternative 3:

- Add concentrated sulfur acid and a carboxylic acid; [1 mark]
- Ester smell; [1 mark]

Alternative 4:

- Add sodium; [1 mark]
- Bubbles / effervescence; [1 mark]

[Total: 2 marks]

- **Careful:** The test you describe must be for an alcohol
- You cannot use bromine water as this tests for an unsaturated compound (the presence of a double bond)
- Both leaf alcohol, and the alkene formed from dehydration have double bonds

4a

a)

i) The molecular formula of 2-ethyl-3-methylbutanoic acid is:

- $\text{C}_7\text{H}_{14}\text{O}_2$; [1 mark]

ii) The number of chiral carbon atoms present in a molecule of 2-ethyl-3-methylbutanoic acid is:

- One; [1 mark]

- Add concentrated sulfur acid and a carboxylic acid; [1 mark]
- Ester smell; [1 mark]

Alternative 4:

- Add sodium; [1 mark]
- Bubbles / effervescence; [1 mark]

[Total: 2 marks]

- **Careful:** The test you describe must be for an alcohol
- You cannot use bromine water as this tests for an unsaturated compound (the presence of a double bond)
- Both leaf alcohol, and the alkene formed from dehydration have double bonds

4a

a)

i) The molecular formula of 2-ethyl-3-methylbutanoic acid is:

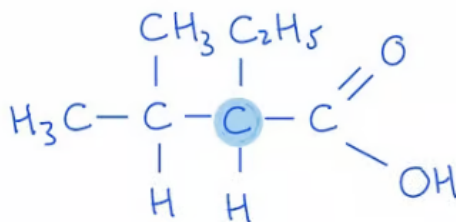
- $C_7H_{14}O_2$; [1 mark]

ii) The number of chiral carbon atoms present in a molecule of 2-ethyl-3-methylbutanoic acid is:

- One; [1 mark]

[Total: 2 marks]

- A chiral carbon is one attached to four different atoms / groups of atoms
- It is useful to draw the displayed formula for the structure so you can identify chiral carbons more clearly:



4b

b)

i) The reagents used and the colour change that would be seen is:

Alternative 1:

- Acidified $K_2Cr_2O_7$ / acidified potassium dichromate (VII); [1 mark]
- From orange; [1 mark]
- To green; [1 mark]

Alternative 2:

- Acidified $KMnO_4$ / potassium manganate (VI); [1 mark]
- From purple; [1 mark]
- To colourless; [1 mark]

ii) The main organic impurity present in the sample of the acid could be:

- 2-ethyl-3-methylbutanal / $(CH_3)_2CHCH(C_2H_5)CHO$; [1 mark]
- Because partial oxidation of alcohol will form an aldehyde; [1 mark]

iii) The apparatus used should be:

- Reflux

AND

Because the alcohol must be fully oxidised; [1 mark]

[Total: 6 marks]

- **Remember:** The oxidising agents used to convert the alcohol to a carboxylic acid, are themselves reduced which results in the colour changes
- Questions about these colour changes occur frequently so make sure you learn them the right way around

4c

c) The colour change you would see if this were heated with the reagents you have given in (b)(i) is:

- None / no colour change; [1 mark]
- (Because) the alcohol is tertiary; [1 mark]

- So cannot be oxidised; [1 mark]

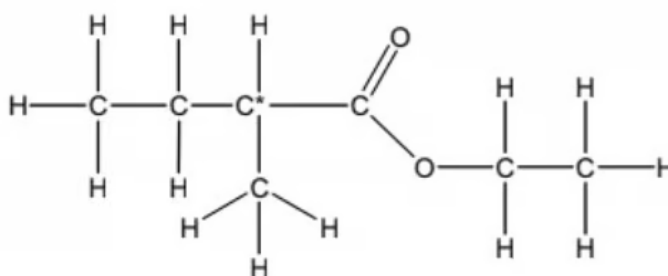
[Total: 3 marks]

- The -OH group is bonded to a carbon atom that is bonded to three other alkyl groups so it is a tertiary alcohol

4d

d)

i) The displayed formula of this ethyl ester and chiral carbon atom is:



- Correct structure; [1 mark]
- Bonds between all atoms shown; [1 mark]
- Correct chiral C atom identified; [1 mark]

[Total: 3 marks]

- The molecular formula of 2-ethyl-3-methylbutanoic acid is $C_7H_{14}O_2$
- Start by drawing the -COO- group
- You are told that it is an ethyl ester, so the chain connected to the oxygen in the -COO- group is C_2H_5
- This leaves four carbons and nine hydrogen atoms
- You know that the remaining four carbons should not be straight chained, but branched
- Draw a chain of three carbon atoms on the left hand side of your ester, and play around moving the location of the methyl group until it is in a position that there is a chiral carbon
- You can achieve marking point 1 if you have not shown the bonds between all atoms
- If you have made an error in drawing the structure, you can still score mark 3 for identifying a correct chiral carbon